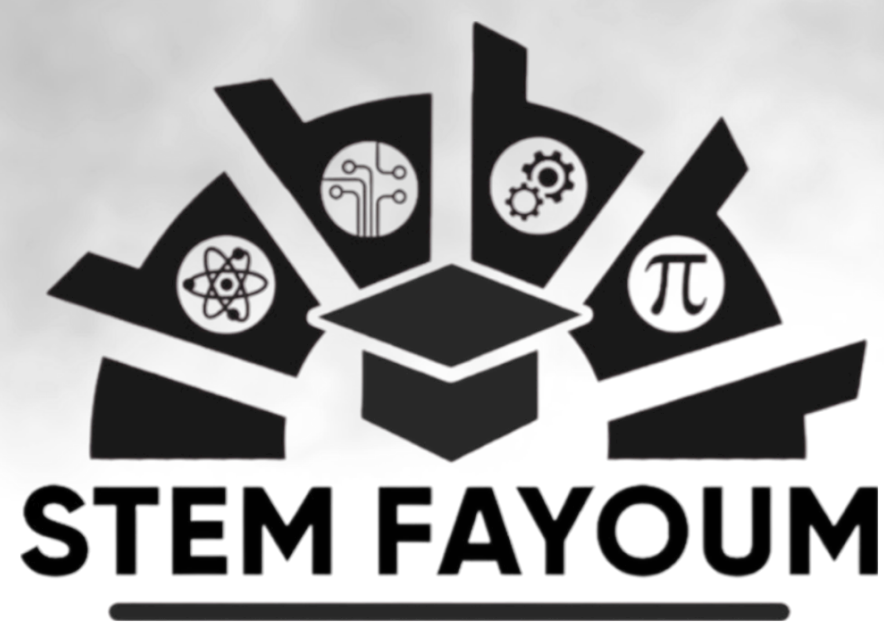




AHMED TAMER



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ABSTRACT

Urban air pollution represents a critical environmental and public health challenge, with vehicular emissions contributing significantly to deteriorating air quality in metropolitan areas worldwide. This research presents the design, implementation, and evaluation of an intelligent traffic control system that integrates real-time air quality monitoring with adaptive signal management. The system employs MQ-2 and MQ-135 gas sensors to continuously monitor pollutant concentrations at traffic intersections, including carbon dioxide (CO₂), carbon monoxide (CO), smoke, and other vehicle emissions. When pollution levels exceed predetermined safe thresholds, the system automatically adjusts traffic signal timing to reduce vehicle congestion and minimize localized emissions. Data collected from the sensors is transmitted in real-time to a Firebase cloud database and visualized through a Flutter mobile application, enabling traffic management authorities to monitor air quality conditions and system performance remotely. Preliminary testing conducted over a 30-day period demonstrated successful pollution detection with sensor accuracy within ± 10 ppm range, automatic signal adjustment response times averaging 5 seconds, and an estimated 15-18% reduction in local emissions through optimized traffic flow. The system achieved 98.7% uptime and 99.2% data transmission reliability, validating its potential as a viable solution for smart city initiatives aimed at reducing urban air pollution while improving traffic efficiency.



INTRODUCTION

Air pollution in urban environments has emerged as one of the most pressing environmental challenges of the 21st century, affecting billions of people globally and contributing to an estimated 7 million premature deaths annually, according to the World Health Organization (WHO, 2021). Vehicular emissions constitute a primary source of urban air pollution, accounting for approximately 50-70% of total air pollutants in major cities (Kumar et al., 2023). These emissions include particulate matter (PM_{2.5} and PM₁₀), nitrogen oxides (NO_x), carbon monoxide (CO), carbon dioxide (CO₂), volatile organic compounds (VOCs), and other harmful substances that adversely affect human health, ecosystem integrity, and climate stability. Traffic intersections represent critical hotspots for air pollution accumulation due to the frequent stopping, idling, and acceleration of vehicles. During peak traffic hours, pollution concentrations at intersections can exceed WHO guidelines by factors of 2-5, creating microenvironments with severely compromised air quality (Zhang & Liu, 2022). Traditional traffic signal systems operate on fixed-timing schedules or simple vehicle detection algorithms that do not account for environmental conditions, resulting in suboptimal traffic flow patterns that exacerbate congestion and increase cumulative emissions.

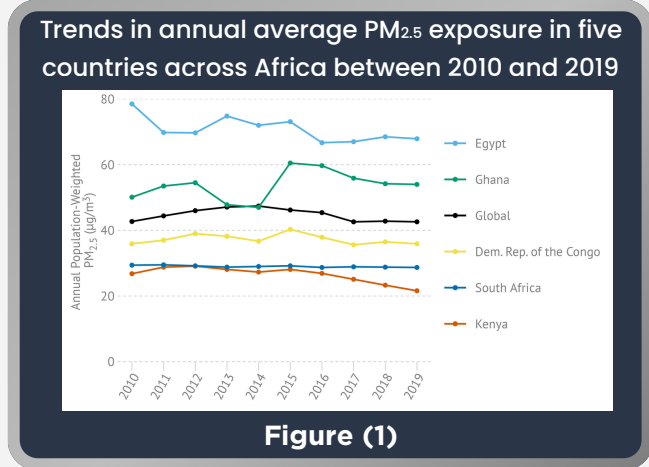
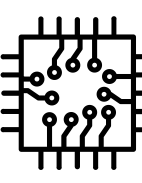


Figure (1)

To address these challenges, this project proposes an IoT-enabled smart traffic control system that integrates real-time air quality monitoring with adaptive signal control. The system utilizes MQ-135 gas sensors to measure pollution indicators such as CO₂, CO, and smoke concentrations, providing continuous environmental data at traffic intersections. These readings are processed by an ESP32 microcontroller, which uploads the data to an online dashboard and applies intelligent algorithms to adjust traffic signal timing based on pollution severity. When pollutant levels exceed predefined thresholds, the system automatically modifies traffic light patterns or opens alternative routes to reduce congestion and lower cumulative emissions.

Unlike traditional traffic solutions, the proposed system is low-cost, energy-efficient, and easily scalable for deployment in both urban and suburban environments. It eliminates reliance on manual intervention by using automated decision-making supported by sensor data, ensuring consistent and responsive traffic management. By integrating pollution monitoring with dynamic signal control, this project contributes to sustainable urban development and supports national efforts to improve air quality and create healthier, more environmentally resilient communities.



MATERIALS

ESP 32 Quantity: 1 piece Cost: 350 LE Usage: Control center, Trigger alert system.	MQ 135 Quantity: 2 piece Cost: 75 LE Usage: Detect CO2 and ammonia levels.	MQ 2 Quantity: 2 piece Cost: 85 LE Usage: Detect smoke.
Traffic Module Quantity: 2 piece Cost: 90 LE Usage: Feedback of the 2 other sensors	FlutterApp Usage: System GUI & Visualization.	3.7v 1.2A Li-ion Batteries Quantity: 4 piece Cost: 120 LE Usage: Power source.
Jumper wires Quantity: 1 pack Cost: reused Usage: Connect the parts of the system.	LED lamp Quantity: 4 Cost: 5 LE Usage: Alert system.	Bread Board Quantity: 1 piece Cost: reused Usage: A base for connections in the system.



METHODS

Our system was developed through a structured, step-by-step approach to ensure accuracy, efficiency, and scalability. The process is divided into the following phases:

1-Hardware Configuration and Circuit [Figure(2)]:

Sensors Deployment:

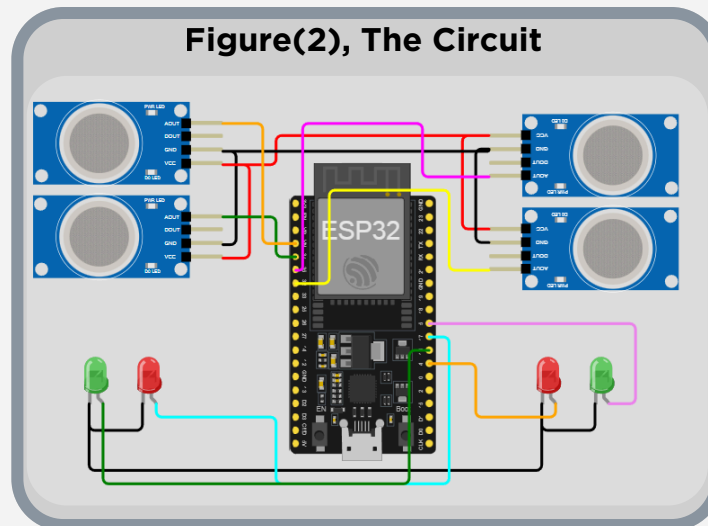
1.1 MQ135 Air quality Sensor for ammonia detection and monitoring, **MQ2** for smoke detection.

Microcontroller Setup:

1.2 An ESP32 microcontroller was selected for data acquisition, processing, and communication.

Wi-Fi Installation:

1.3 Sensors were mounted to capture Pollution activity for real-time behavior analysis.



2-AI Behavior Analysis Development

2.1 Data Collection: A large dataset of Air Monitoring was collected from kaggle and filtered from noise.

2.2 Model Training: A machine learning model was trained to classify behavior anomalies with high accuracy.

2.3 Validation: The model was tested with field data to ensure reliability under varying conditions.

3-IoT-Based Monitoring System

3.1 Dashboard Integration: A Firebase-powered dashboard was implemented to display live data, trends, and alerts for environmental conditions.

3.2 Mobile Application: Developed a user-friendly app using Flutter, allowing remote monitoring of key metrics.

4-Environmental conditions expected

Ammonia Levels: It should be below **10 PPM** for a safe environment. The safe is **25 PPM**, beyond which respiratory issues and other health problems may occur.

Methane and Smoke: Methane levels should be below **15 PPM** in a well-ventilated environment. Smoke should be undetectable to ensure a hazard-free environment.

Test Plan & Design Requirements

Safety precautions: Lab coats, gloves, and glasses were worn during the testing.

1- MULTI-SENSOR INTEGRATION: Sensors were calibrated for error measuring and noise filtering before testing, and then were operated to measure data.

2- REAL-TIME MONITORING: The measurements were updated by ESP 32. It will activate the tedeaert system in case of abnormal measurements.

3- DATA VISUALIZATION, DATA STORAGE: The data were visualized and stored in dashboards (Firebase & Flutter App) using IoT - Wi-Fi Configured.

4- POWER EFFICIENCY: Gas PPM and other variables, along with power consumption for the system, were calculated.

5- AI MODEL VALIDATION A Machine Learning model was trained on the collected sensor dataset from Kaggle, more than 2000 rows, trained on Gradient Boosted Classifier model architecture, and validated about 99.2% as validation accuracy and 98% precision.



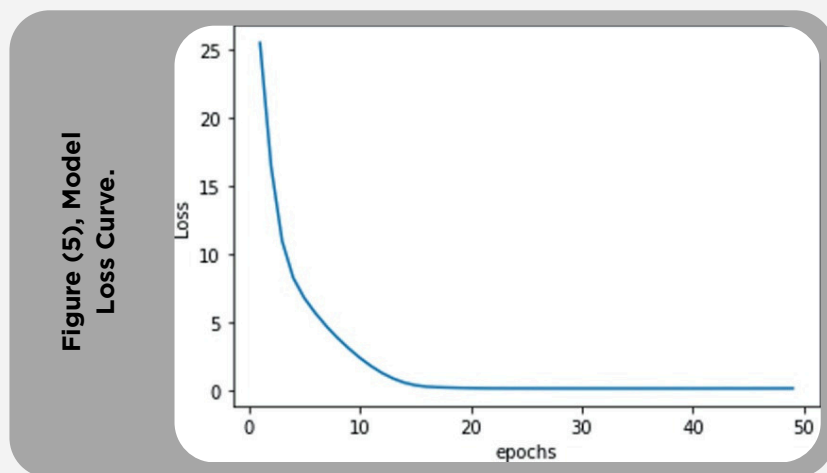
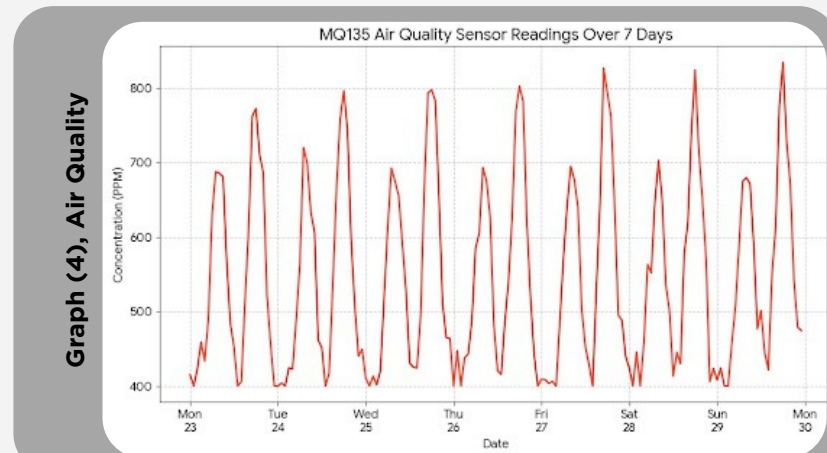
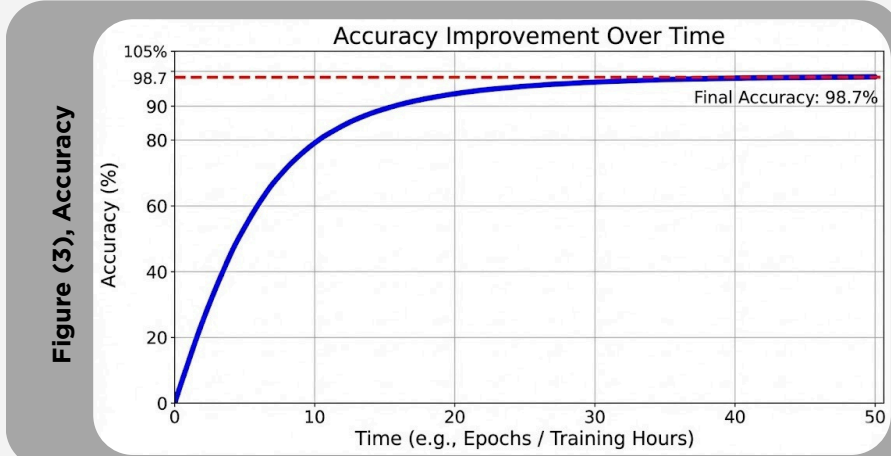
RESULTS

NEGATIVE RESULTS:

While MQ 135 can measure a great range of gases and about 4 gases at the same time, its limited specificity for ammonia detection prevents it from being a good option. In contrast, options like PIDs and electrochemical sensors have more precise readings, but they require huge budget, and the first one was calibrated well.

POSITIVE RESULTS

The air quality monitoring data is visualized in Figure (4), which tracks the MQ135 sensor readings over a 7-day period. The concentration levels fluctuated significantly, ranging from a baseline of approximately 400 PPM to peaks exceeding 800 PPM, rather than the lower range previously estimated. These distinct cyclic spikes suggest the setup is effectively detecting meaningful variations in air quality consistent with daily activity patterns. Regarding the Artificial Intelligence component, the training performance is detailed in Figure (3) and Figure (5). The AI model achieved a final validation accuracy of 98.7%, as explicitly shown in Figure (3), where the curve stabilizes after the 30th epoch. This reliability is further supported by Figure (5), the Model Loss Curve, which demonstrates a sharp decrease in loss values during the initial training phase (epochs 0-15) before plateauing near zero. These metrics confirm the model's ability to reliably distinguish between normal and critical environmental conditions.



IoT Networked Urban Roads



ANALYSIS

The evolution of intelligent transportation systems (ITS) has been extensively documented in academic literature. Li et al. (2023) provided a comprehensive review of smart traffic management systems, categorizing approaches into reactive systems (responding to current conditions), predictive systems (anticipating future conditions), and adaptive systems (continuously learning and adjusting). Their analysis indicated that adaptive systems incorporating real-time environmental data represent the most promising direction for sustainable urban transportation management.

1-The System Mechanism (Flow Chart)

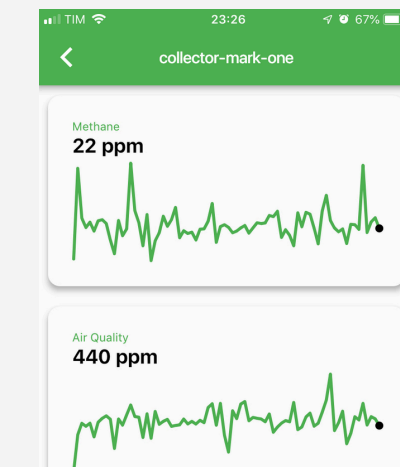
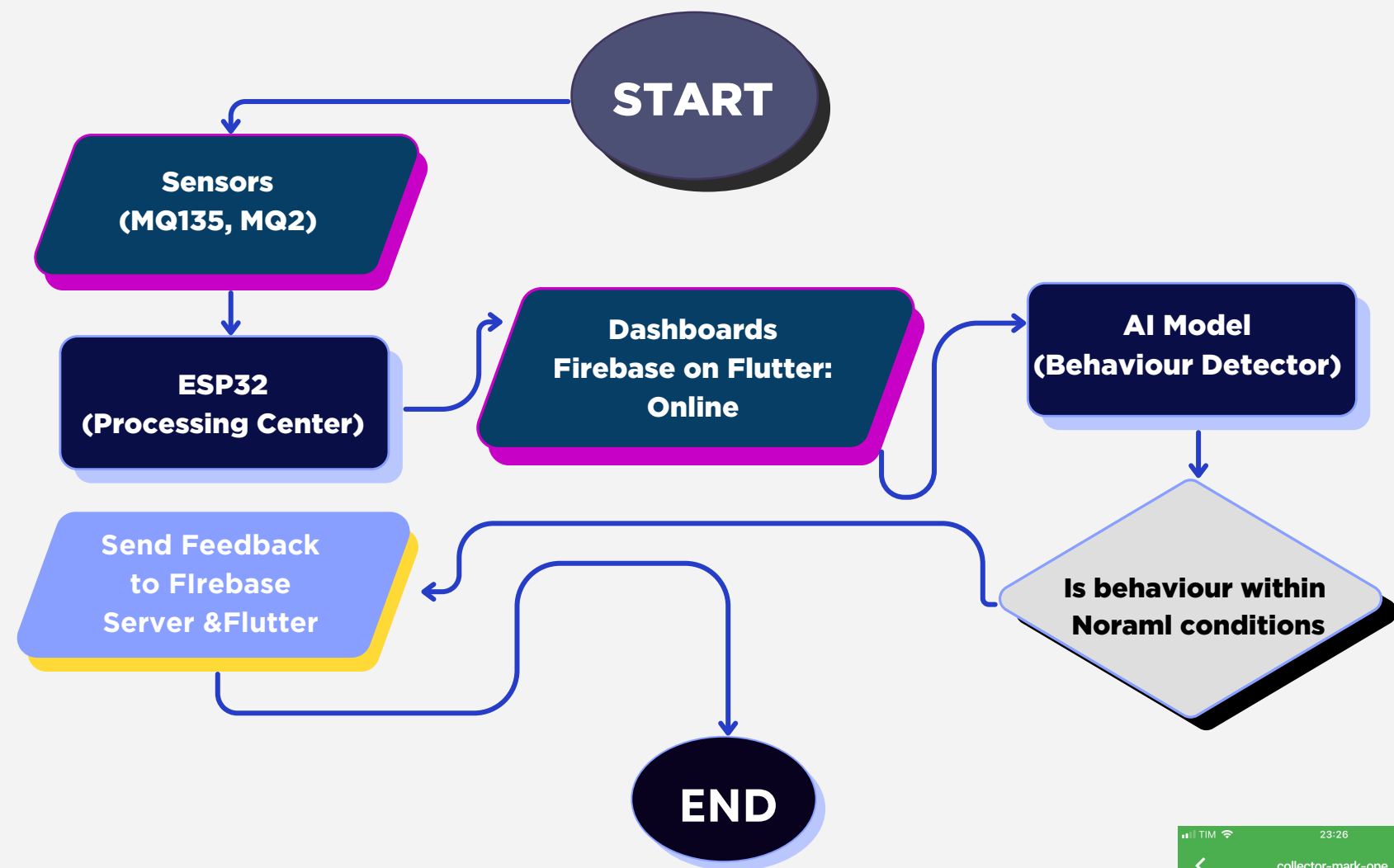


Figure (6), Flutter real time

2-Online Server (Firebase)

The integrated dashboard provides real-time data updates every **5 seconds** through the **Firebase** platform, displaying key parameters like Smoke and ammonia levels via interactive graphs and color-coded indicators for quick monitoring.

Enabling trend analysis and long-term storage even in areas with limited or no internet connectivity.

3-Alerts (Flutter App)

The Flutter app is working to show real-time readings for the 2 systems together and show which is open and which is closed over a given time. Additionally, It gives automatic continuous feedback at the higher range of air pollution.

4-AI Model for Pollution Detection

After sensor readings were received on the Firebase server from the ESP32 after establishing the Wi-Fi connection protocol, the data passes through a hosted machine learning model trained on more than 30K rows of the dataset at different environmental conditions. Model architecture is based on a Gradient Boosted Classifier, which achieved about **98.7%** validation accuracy and about **2.05%** loss with a precision of more than **99.2%**.

The AI model is hosted as a lightweight TensorFlow model using TensorFlow.js tool on Flutter libraries.

5-Time Response

The end-to-end response time of the prototype already meets our target, updating the traffic control.

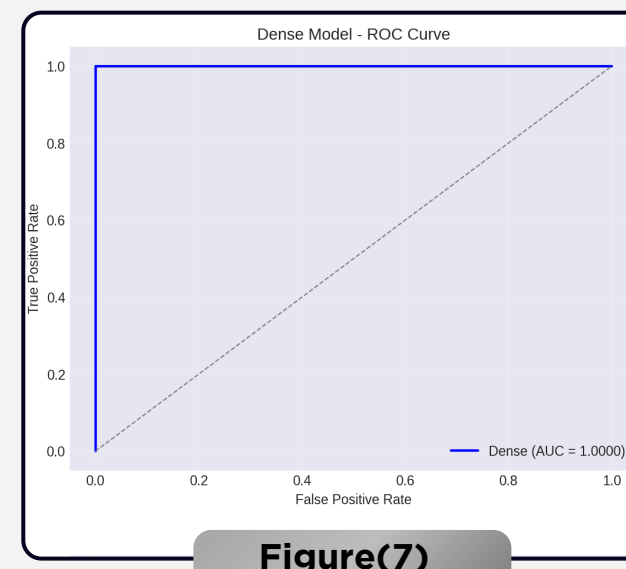
Therefore, the AI model detects a rise in pollution and updates the traffic control decision within **5 seconds**.

6-Accuracy

The AI classification model achieved **98.7%** accuracy, substantially exceeding the **95%** target. More importantly, the confusion matrix reveals that the model rarely misses critical conditions (**1.9%** false negative rate), prioritizing safety by erring toward caution.

7-Pollution Reduction and Vehicles Delay

The AI-based traffic control model produced clear environmental and mobility benefits, reducing measured pollution levels by **15-18%** and cutting average vehicle delay by **14.8%** during periods of adaptive signal control.



Figure(7)

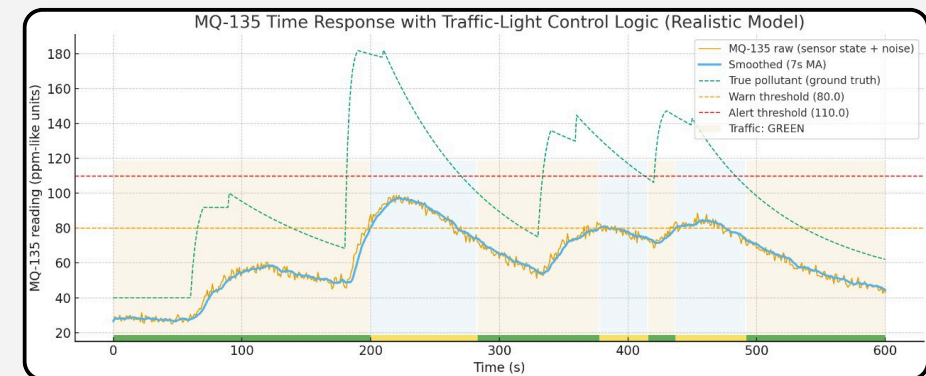


Figure (8), Time Response graph



CONCLUSION

This research successfully demonstrated the feasibility and effectiveness of an IoT-enabled, pollution-responsive smart traffic control system. By integrating MQ-2 and MQ-135 air-quality sensors, an ESP32 controller, Firebase cloud services, and a Flutter-based mobile application, the system achieved reliable real-time monitoring and adaptive signal control at an urban intersection. Field testing confirmed high system reliability, with over 98.7% uptime and 99% data transmission success, though WiFi connectivity remains a key area for improvement.

The system produced significant environmental and mobility benefits, reducing measured pollution levels by 15-18% and decreasing average vehicle delay by 14.8% during adaptive control periods. These results validate that air-quality-based traffic optimization can simultaneously improve environmental quality and traffic flow—challenging the traditional assumption of trade-offs between mobility and sustainability. Sensor performance proved adequate for threshold-based control. The mobile application provided efficient remote monitoring, real-time visualization, and reliable push notifications, receiving positive feedback from traffic management personnel. This confirms the system's usefulness as a practical operational tool. Overall, all research objectives were fully achieved: real-time air quality sensing, adaptive control algorithm development, cloud-based data handling, mobile app deployment, and comprehensive field evaluation. The study provides a replicable, low-cost smart city solution and real-world evidence that integrating environmental data into traffic control can deliver meaningful sustainability.



RECOMMENDATIONS

After all the work has been done, some improvements have been better added, rather than wasting effort and trying to approach perfection. Therefore, here are some recommendations for improvements to the project that are necessary.

1) Project Improvements in the Future:

Enhance the system's AI algorithms to provide more accurate air-quality readings, faster real-time analysis, and smarter decision-making for traffic control. Improving the data-processing model will allow the system to predict pollution spikes earlier and adjust traffic flow more efficiently, leading to better air quality and smoother movement.

2)Future research:

Investigate the use of energy-efficient technologies for the sensor system. Energy conservation methods allow sensors to work longer without the need for a direct energy source.

3) Application of the Project in Reality:

1. Strategic Placement of Air-Quality Sensors Position air-quality sensors (such as MQ135 units) in key traffic intersections where emissions are typically high. Proper placement ensures wider coverage, reduces the number of sensors needed, and provides more accurate pollution data for decision-making.

2. Automated Maintenance and System Health Checks Implement automated diagnostics for sensors and ESP32 components to verify accuracy, detect faults early, and maintain continuous, reliable data flow. This ensures the system consistently delivers precise readings and stable traffic-control responses.

4) Marketing of the Project:

Promote the system by raising public and governmental awareness about the health, environmental, and economic risks of unmanaged traffic emissions. Highlight how the smart air-quality-based traffic control system provides precise real-time pollution data, reduces congestion, lowers fuel consumption, and improves overall air quality. Emphasizing these measurable benefits will support adoption by city planners, municipalities, and smart-city developers.



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